

$$\text{tuple}(X_1, \dots, X_n) = X^n = X^{1:n}$$

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(I) : Induction

Conditioning on X

$$d_{TV}(P, Q) = \frac{1}{2} \sum_{x^n \in X^n} |P(x^n) - Q(x^n)| = \frac{1}{2} \sum_{x^n \in X^n} \left| \sum_{x^{1:n-1} \in X^{1:n-1}} P(x^n | x^{1:n-1}) P(x^{1:n-1}) - Q(x^n | x^{1:n-1}) Q(x^{1:n-1}) \right|$$

by Triangle Inequality

$$\leq \frac{1}{2} \sum_{x^n \in X^n} \left| \sum_{x^{1:n-1} \in X^{1:n-1}} P(x^n | x^{1:n-1}) P(x^{1:n-1}) - P(x^{n-1}) Q(x^n | x^{1:n-1}) + P(x^{n-1}) Q(x^n | x^{1:n-1}) - Q(x^{1:n} | x^{1:n-1}) Q(x^{1:n-1}) \right|$$

$$\leq \frac{1}{2} \sum_{x^n \in X^n} \left| \sum_{x^{1:n-1} \in X^{1:n-1}} P(x^n) (P(x^n | x^{n-1}) - Q(x^n | x^{n-1})) + \sum_{x^{1:n-1} \in X^{1:n-1}} Q(x^n | x^{n-1}) (P(x^{n-1}) - Q(x^{n-1})) \right|$$

$$= \frac{1}{2} \sum_{x^n \in X^n} \sum_{x^{1:n-1} \in X^{1:n-1}} P(x^n) |P(x^n | x^{n-1}) - Q(x^n | x^{n-1})| + \frac{1}{2} \sum_{x^n \in X^n} \sum_{x^{1:n-1} \in X^{1:n-1}} Q(x^n | x^{n-1}) |P(x^{n-1}) - Q(x^{n-1})|$$

$= \mathbb{E} Q(\cdot | \cdot) + \mathbb{E} Q(\cdot | \cdot)$

$$= \mathbb{E}_{x^{n-1} \sim P} \left[\|P(\cdot | x^{n-1}) - Q(\cdot | x^{n-1})\|_{TV} \right] + \frac{1}{2} \sum_{x^{n-1} \in X^{n-1}} |P(x^{n-1}) - Q(x^{n-1})|$$

$$= \mathbb{E}_{x^{n-1} \sim P} \left[\|P(\cdot | x^{n-1}) - Q(\cdot | x^{n-1})\|_{TV} \right] + \|P(x^{n-1}) - Q(x^{n-1})\|_{TV}$$

$\hookrightarrow P^{n-1} - Q^{n-1}$

so now we proved that: $\|P^n - Q^n\|_{TV} \geq \|P^{n-1} - Q^{n-1}\|_{TV} + \mathbb{E}_{x^{n-1} \sim P^{n-1}} [\|P(\cdot | x^{n-1}) - Q(\cdot | x^{n-1})\|_{TV}]$
 by applying this n times we can deduce that:

$$\|P - Q\|_{TV} \geq d_{TV}(P, Q) \leq \sum_{i=1}^n \mathbb{E}_{x^{i-1} \sim p_{i-1}} [\|P_i(\cdot | x_{1:i-1}) - Q_i(\cdot | x_{1:i-1})\|_{TV}] \quad \underline{Q.E.D.}$$

II $\|P - Q\|_{TV} = \frac{1}{2} \sum_{x \in X} |P(x) - Q(x)|$

$$\left. \begin{aligned} 1 + TV(P, Q) &= \sum_{x \in X} \max(P(x), Q(x)) \\ 1 - TV(P, Q) &= \sum_{x \in X} \min(P(x), Q(x)) \end{aligned} \right\} \quad 1 - TV^2(P, Q) = \left(\sum_{x \in X} \max\{P(x), Q(x)\} \right) \left(\sum_{x \in X} \min\{P(x), Q(x)\} \right)$$

By Cauchy Schwarz

$$1 - TV^2(P, Q) \geq \left(\sum_{x \in X} \sqrt{\min(P(x), Q(x)) \cdot \max(P(x), Q(x))} \right)^2 \geq \mathcal{B}^2(P, Q) \Rightarrow \begin{cases} 1 - TV^2 \geq \mathcal{B}^2 \\ TV \leq \sqrt{1 - \mathcal{B}^2} \end{cases}$$

now using Jensen inequality we obtain an upper bound for $BC(P, Q)$:

$$BC(P, Q) = \mathbb{E}_{x \sim Q} \left[\sqrt{\frac{p(x)}{q(x)}} \right] = \mathbb{E} \left[e^{\frac{1}{2} \ln \frac{p(x)}{q(x)}} \right] \stackrel{(*)}{\geq} e^{\frac{1}{2} \mathbb{E} \ln \left[\frac{p(x)}{q(x)} \right]} = e^{-\frac{1}{2} D_{KL}(P, Q)}$$

$$\stackrel{(*)}{\Rightarrow} BC(P, Q) \geq e^{-\frac{1}{2} D_{KL}(P, Q)} \quad \stackrel{(*)}{\Rightarrow} TV \leq \sqrt{1 - BC^2} \leq \sqrt{1 - e^{-D_{KL}(P, Q)}}$$

Now for the last part we take advantage of convex function's properties:

let $f(x) = \sqrt{1-x}$ $\xrightarrow{Cvx} f(x) \leq f(y) + (x-y)f'(y) \xrightarrow{y=0} f(x) \leq f(0) + \lambda f'(0) = 1 - \frac{x}{2}$

$$x = e^{-D_{KL}(P, Q)} \Rightarrow \sqrt{1 - e^{-D_{KL}(P, Q)}} \leq 1 - \frac{1}{2} D_{KL}(P, Q)$$

$\Rightarrow$ Now we proved all the inequalities. Q.E.D

$$TV \leq \sqrt{1 - BC^2} \leq \sqrt{1 - e^{-D_{KL}(P, Q)}} \leq 1 - \frac{1}{2} D_{KL}(P, Q)$$

III) since $BC(P_n^{\otimes n}, Q_n^{\otimes n}) \stackrel{iid}{=} BC(P_n, Q_n)^n$ & $BC = 1 - H^2$ & the fact that $1 - BC(P_n, Q_n)^n \leq TV(P_n^{\otimes n}, Q_n^{\otimes n}) \leq \sqrt{1 - BC(P_n, Q_n)^{2n}}$ (4). Let $a_n = n H^2(P_n, Q_n)$

then $BC(P_n, Q_n) = 1 - \frac{a_n}{n} \rightarrow$ plugging in (4) implies that:

$$1 - \left(1 - \frac{a_n}{n}\right)^n \leq TV(P_n^{\otimes n}, Q_n^{\otimes n}) \leq \sqrt{1 - \left(1 - \frac{a_n}{n}\right)^{2n}}$$

since $\lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k$ we can deduce that:

$$\lim_{n \rightarrow \infty} 1 - \left(1 - \frac{a_n}{n}\right)^n \leq \lim_{n \rightarrow \infty} TV(P_n^{\otimes n}, Q_n^{\otimes n}) \leq \lim_{n \rightarrow \infty} \sqrt{1 - \left(1 - \frac{a_n}{n}\right)^{2n}}$$

$$\lim_{n \rightarrow \infty} 1 - e^{-a_n} \leq \lim_{n \rightarrow \infty} TV(P_n^{\otimes n}, Q_n^{\otimes n}) \leq \lim_{n \rightarrow \infty} \sqrt{1 - e^{-2a_n}}$$

now if $\begin{cases} \lim_{n \rightarrow \infty} a_n = 0, \text{ then } \lim_{n \rightarrow \infty} 1 - e^{-a_n} = 1 - 1 = 0 \\ \lim_{n \rightarrow \infty} a_n = \infty, \text{ then } \lim_{n \rightarrow \infty} 1 - e^{-a_n} = 1 - 0 = 1 \end{cases} \quad (***)$

if $\begin{cases} \lim_{n \rightarrow \infty} a_n = 0, \text{ then } \lim_{n \rightarrow \infty} \sqrt{1 - e^{-2a_n}} = 0 \\ \lim_{n \rightarrow \infty} a_n = \infty, \text{ then } \lim_{n \rightarrow \infty} \sqrt{1 - e^{-2a_n}} = 1 \end{cases} \quad (****)$

argmaxing (ψ_k) , (ψ_k) results in $\begin{cases} \text{TV}(P_n^{\otimes n}, Q_n^{\otimes n}) = 0 & \text{if } a_n \rightarrow 0 \quad (\text{I}) \\ \text{TV}(P_n^{\otimes n}, Q_n^{\otimes n}) = 1 & \text{if } a_n \rightarrow \infty \quad (\text{II}) \end{cases}$

(اسم انگریزی (نہایت) سے) $\rightarrow$ $\begin{cases} \text{TV}(P_n^{\otimes n}, Q_n^{\otimes n}) = 0 & \text{if } a_n \rightarrow 0 \quad (\text{I}) \\ \text{TV}(P_n^{\otimes n}, Q_n^{\otimes n}) = 1 & \text{if } a_n \rightarrow \infty \quad (\text{II}) \end{cases}$

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(I) $a_n \rightarrow 0 \Rightarrow H^2 = \frac{a_n}{n} \Rightarrow H^2 \approx O(\frac{1}{n})$

(II) $a_n \rightarrow \infty \Rightarrow H^2 = \frac{a_n}{n} \Rightarrow H^2 \approx O(\frac{1}{n})$

so at last we can infer that:

$\begin{cases} a_n \rightarrow 0 \Rightarrow (\text{dTV}(P_n^{\otimes n}, Q_n^{\otimes n}) \rightarrow 0) \Leftrightarrow (D_{H^2}(P_n, Q_n) \propto \frac{1}{n}) \\ a_n \rightarrow \infty \Rightarrow (\text{dTV}(P_n^{\otimes n}, Q_n^{\otimes n}) \rightarrow 1) \Leftrightarrow (D_{H^2}(P_n, Q_n) \propto \frac{1}{n}) \end{cases}$

IV same $\int_{x \in \mathcal{X}} q(x) dx = 1 \Rightarrow \frac{1}{2} \sqrt{\int_{\mathcal{X}} \frac{(P_1(x) - P_2(x))^2}{q(x)} dx} = \frac{1}{2} \sqrt{\int_{\mathcal{X}} \frac{(P_1(x) - P_2(x))^2}{q(x)} dx} \int_{\mathcal{X}} q(x) dx$

$\forall q \in \mathcal{P} \Rightarrow \frac{1}{2} \sqrt{\int_{\mathcal{X}} \frac{(P_1(x) - P_2(x))^2}{q(x)} dx} \geq \frac{1}{2} \int_{\mathcal{X}} \sqrt{(P_1(x) - P_2(x))^2} dx = \int_{\mathcal{X}} |P_1(x) - P_2(x)| dx = \text{TV}(P_1, P_2)$

Cauchy Schwartz

Now we will investigate the case of equality here, we know that equality holds in Cauchy-Schwarz when all of the terms have the same ratio:

$\forall x \Rightarrow \frac{|P_1(x) - P_2(x)|}{q(x)} = c \xrightarrow{\int q = 1} q(x) = \frac{|P_1(x) - P_2(x)|}{2 \text{TV}(P_1, P_2)}$

so we can now deduce that

$\text{TV}(P, Q) = \frac{1}{2} \inf_{q \in \mathcal{P}} \sqrt{\int_{\mathcal{X}} \frac{(P(x) - Q(x))^2}{q(x)} dx}$

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V $\begin{matrix} & & P_{Y|X} & & \\ & \nearrow & & \searrow & \\ 0 & X & \xrightarrow{P_{Y|X}} & Y & \\ & \nwarrow & & \nearrow & \\ 1 & & P_0: P_{Y|X=0} & & \\ & & P_1: P_{Y|X=1} & & \end{matrix}$

let $\begin{cases} P[X=0] = p \\ P[X=1] = \bar{p} \end{cases}$

$P_Y = p P_0 + \bar{p} P_1$

$I(X; Y) = \mathbb{E}_X [D_{KL}(P_{Y|X} \| P_Y)] = p D_{KL}(P_0 \| P_Y) + \bar{p} D_{KL}(P_1 \| P_Y) \quad (*)$

now we'll use Pinsker to relate it to TV

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$$\begin{cases} \text{TV}(P_0 \| P_4) = \text{TV}(P_0 \| P P_0 + \bar{P} P_1) = \frac{1}{2} \sum_{x \in \mathcal{X}} \left| \frac{P_0(x) - P P_0(x) - \bar{P} P_1(x)}{\bar{P} P_0(x)} \right| = \bar{P} \text{TV}(P_0 \| P_1) \\ \text{TV}(P_1 \| P_4) = \text{TV}(P_1 \| P P_0 + \bar{P} P_1) = \frac{1}{2} \sum_{x \in \mathcal{X}} \left| \frac{P_1(x) - P P_0(x) - \bar{P} P_1(x)}{P P_0(x)} \right| = P \text{TV}(P_0 \| P_1) \end{cases}$$

now Pinsker tells us that

$$\text{TV}(P \| Q) \leq \sqrt{\frac{1}{2} D_{KL}(P \| Q)}$$

$\Rightarrow$

$$I(X; Y) = P D_{KL}(P_0 \| P_4) + \bar{P} D_{KL}(P_1 \| P_4) \geq 2P \text{TV}^2(P_0 \| P_4) + 2\bar{P} \text{TV}^2(P_1 \| P_4)$$

$$\Rightarrow I(X; Y) \geq 2P\bar{P}^2 \text{TV}^2(P_0, P_1) + 2\bar{P}P^2 \text{TV}^2(P_1, P_0) = 2P\bar{P} \text{TV}^2(P_0, P_1) (P + \bar{P})$$

now if we have a BSC $\rightarrow P = \frac{1}{2}$

$$\Rightarrow \boxed{I(X; Y) \geq \frac{1}{2} \text{TV}^2(P_0, P_1)}$$

4 now for the other side of inequality

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Now we'll prove the other side of inequality which is $I(X;Y) \leq d_{TV}(P_0, P_1)$

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(V) sol

Since $d_{KL} \leq d_{TV} \rightarrow \chi^2(P||Q) \geq D_{KL}(P||Q)$, therefore:

$$\begin{cases} \chi^2(P_0||P_1) = \sum_{x \in \mathcal{X}} \frac{(P_0 \cdot \bar{P} + P_1 \cdot \bar{P})^2}{P_0 \cdot \bar{P} + \bar{P} \cdot P_1} = \bar{P}^2 \sum_{x \in \mathcal{X}} \frac{(P_0 - P_1)^2}{P_0 + (P_0 - P_1) \cdot \bar{P}} \\ \chi^2(P_1||P_0) = \sum_{x \in \mathcal{X}} \frac{(P_1 \cdot P_0 - P_1 \cdot \bar{P})^2}{P_1 \cdot P_0 + \bar{P} \cdot P_1} = P^2 \sum_{x \in \mathcal{X}} \frac{(P_0 - P_1)^2}{P_1 \cdot P_0 + \bar{P} \cdot P_1} \end{cases}$$

$$\begin{aligned} I(X;Y) &\leq D_{KL}(P_0||P_1) \leq P \cdot D_{KL}(P_0||P_1) + \bar{P} \cdot D_{KL}(P_1||P_0) \\ &\leq P \chi^2(P_0||P_1) + \bar{P} \chi^2(P_1||P_0) \end{aligned}$$

which implies (since $P\bar{P}^2 + \bar{P}P^2 = P\bar{P}(P+\bar{P}) = P\bar{P}$):

$$I(X;Y) \leq P\bar{P} \sum_{x \in \mathcal{X}} \frac{(P_0 - P_1)^2}{P_1 \cdot P_0 + \bar{P} \cdot P_1} = P\bar{P} \left[\sum_{\substack{x \in \mathcal{X} \\ P_0(x) > P_1(x)}} \dots + \sum_{\substack{x \in \mathcal{X} \\ P_1(x) \leq P_0(x)}} \dots \right]$$

$$(*) \hookrightarrow \leq P\bar{P} \left[\frac{1}{P} \sum_{P_0 > P_1} (P_0 - P_1) + \frac{1}{\bar{P}} \sum_{P_1 > P_0} (P_1 - P_0) \right]$$

$$\text{Since } \begin{cases} \frac{(P_0 - P_1)^2}{P_1 + (P_0 - P_1)\bar{P}} \stackrel{P_0 > P_1}{\leq} \frac{(P_0 - P_1)^2}{P_1 \cdot (P_0 - P_1)} = \frac{P_0 - P_1}{P} \quad (*) \\ \frac{(P_0 - P_1)^2}{P_0 + (P_1 - P_0)\bar{P}} \leq \frac{(P_1 - P_0)^2}{\bar{P}(P_1 - P_0)} = \frac{P_1 - P_0}{\bar{P}} \quad (**) \end{cases}$$

$$\begin{aligned} \Rightarrow I(X;Y) &\leq P \sum_{P_0 > P_1} |P_0 - P_1| + \bar{P} \sum_{P_1 > P_0} |P_1 - P_0| = P d_{TV}(P_0||P_1) + \bar{P} d_{TV}(P_1||P_0) \\ &= d_{TV}(P_0||P_1) \end{aligned}$$

so finally we got the proof:

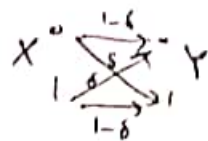
$$\frac{1}{2} d_{TV}^2(P_0||P_1) \leq I(X;Y) \leq d_{TV}(P_0||P_1) \quad \text{Q.E.D.} \blacksquare$$

$G = (V, E)$
vertices v edges e

$$(1) \begin{cases} \{X_v; v \in V\} \stackrel{iid}{\sim} \text{Ber}(1/2) \\ \{Z_e; e = (u,v) \in E\} \stackrel{iid}{\sim} \text{Ber}(\delta) \\ e = (u,v) : Y_e = X_u \oplus X_v \oplus Z_e \end{cases}$$

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(ii) Contraction coefficient for BSC $\rightarrow \eta_{PIX} = (1 - 2\delta)^2$



(iii) $U - X - Y \Rightarrow$
Markov kernel
DPI $I(U; Y) \leq I(U; X)$

$$\eta_{PIX} = \sup_{P_{IX}} \frac{I(U; Y)}{I(U; X)}$$

$$I(U; Y) \leq \eta_{PIX} \cdot I(U; X)$$

(I) $I(X_v; Y_e) = D_{KL}(P_{E|V} \| P_V P_E) = \mathbb{E}_V [D_{KL}(P_{E|V} \| P_E)] = 0$

$$\begin{cases} X_u \sim \text{Ber}(1/2) \\ X_v \sim \text{Ber}(1/2) \\ Z_e \sim \text{Ber}(\delta) \end{cases} \begin{cases} H_{u,v} = X_u \oplus Z_e \rightarrow \mathbb{P}[H_{u,v} = 1] = \delta \cdot \frac{1}{2} + (1-\delta) \cdot \frac{1}{2} = \frac{1}{2} \\ H_{u,v,v} = H_{u,v} \oplus X_v \rightarrow \mathbb{P}[H_{u,v,v} = 1] = 1/2 \end{cases}$$

So we can see that regardless our choices of $X_u, X_v; Y_e \sim \text{Ber}(1/2)$

Z_e can't be determined by X_u, X_v (and generally by the set of $\{X_u; u \in V\}$)

So Y_e can't be determined by $\mathcal{E}$. (Y_e is independent of X_v) in other words.

(II) we will use induction on $|E|$, we want to bound information by knowing X_S for some $S \subseteq V$ and Y_E .

let us assume for the base case that $v \in S \subseteq V$; then $\mathbb{P}[v \rightarrow S] = 1$ & therefore $I(X_v; X_S, Y_E) \leq H(X_v) \leq \log 2 = \mathbb{P}[v \rightarrow S] \cdot \log 2$ $\square$

from now on let $v \notin S, v \in V$. we can easily deduce that, the more v is closer to set S , the more information it'll provide and vice-versa. if there's no path from v to S then $\mathbb{P}[v \rightarrow S] = 0$ & $I(\cdot; \cdot) = 0$.

We start adding edges to see the fully-connected graph. It's obvious that adding an edge which connects two elements of S will not provide us any info. So let $E^{(k)}$ be the set of all edges added adding the k^{th} edge. By conditioning on $E^{(k)}$ for $E^{(k+1)}$ we get:

$$E^{(k+1)} = E^{(k)} \cup \{e\} \quad \text{edge added!}$$

$$I(X_v; X_s, Y_{E^{(k+1)}}) = I(X_v; X_s, Y_{E^{(k)}}) + I(X_v; Y_{E^{(k)}} | X_s, Y_{E^{(k)}})$$

if $e \in S \times S \rightarrow I(X_v, Y_e | X_s, Y_{E^{(k)}}) = 0$ since $Y_e | X_s \perp\!\!\!\perp X_v, Y_{E^{(k)}}$

$$\text{thus: } I(X_v, Y_e | X_s, Y_{E^{(k)}}) = 0$$

So the merely, in case where one of $e = (u, v)$ is at least outside of S , the information will leak, otherwise as we saw no information was leaked.

So by conditioning on $E^{(k)}$ we get:

$$I(X_v, X_s, Y_{E^{(k+1)}}) = I(X_v; X_s, Y_{E^{(k)}}) + I(X_v; Y_e | X_s, Y_{E^{(k)}})$$

& by contraction coefficient we can bound the second term:

$$I(X_v; Y_e | X_s, Y_{E^{(k)}}) \leq \eta_k \cdot I(X_v; X_u | X_s, Y_{E^{(k)}}) \leq H(X_u) = \log 2$$

so by adding each node which is not already in S , we added $\eta \cdot \log 2$ information where $\eta = (1-2\delta)^2$

So now if we average over the set E , we get

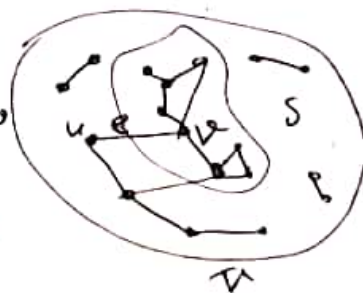
$$I(X_v; Y_e | X_s, Y_{E^{(k)}}) = \mathbb{E}_{e \in E} [I(X_v, Y_e | X_s, Y_{E^{(k)}})] \quad X_u \rightarrow Y_e \leftarrow X_s$$

$$= P[v \rightarrow u] I(X_v, Y_e | X_s, Y_{E^{(k)}}, v \rightarrow u) + P[v \leftarrow u] I(X_v, Y_e | X_s, Y_{E^{(k)}})$$

$$\leq P[v \rightarrow u] \cdot 2 \log 2 = (\log 2) (1-2\delta)^2 P[v \rightarrow u]$$

Now we'll take our inductive step:

$$\rightarrow I(X_v; X_s, Y_{E^{(k+1)}}) \leq I(X_v; X_s, Y_{E^{(k)}}) + \eta P[v \rightarrow u] \log 2 \quad (\text{ok})$$



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since $IP_{G^{k+1}, \eta}[v \rightarrow s] \geq IP[(v \rightarrow s)]_{G^k, \eta} \cup [(v \rightarrow u), (u \rightarrow s)]$

the more general case

$$= IP_{G^k, \eta}[v \rightarrow s] + \eta IP_{G^k, \eta}[v \rightarrow u] \quad (**)$$

(*)
 $\Rightarrow I(X_u; X_s, Y_{\mathcal{E}}^{(k+1)}) \leq \left[IP_{G^k, \eta}[v \rightarrow s] + \eta IP_{G^k, \eta}[v \rightarrow u] \right] \log 2$

$$\leq IP_{G^{k+1}, \eta}[v \rightarrow s] \log 2$$

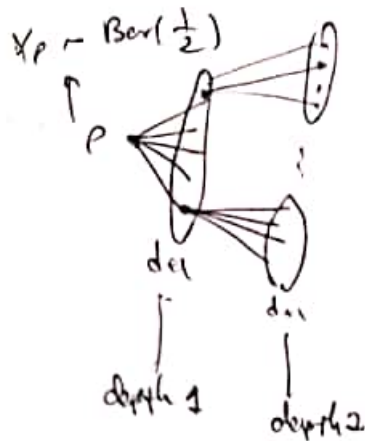
$$\Rightarrow I(X_u; X_s, Y_{\mathcal{E}}) \leq IP_{G^k, \eta}[v \rightarrow s] \log 2$$

using induction into further steps

Q.E.D 

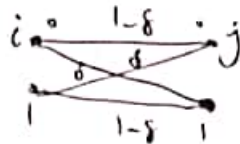
III

We know that in a tree T there's no loop, we have a root r (2 $O(1)$ sets)
 since the tree is plain it looks like this: (III) $\sim 1/2$



$$V \in T \rightarrow |N(i)| = d+1$$

$\forall i, j \in T: N(i) \cap N(j) = \emptyset \rightarrow$ not to form any loop
 each edge is a channel BSC(δ):



S_k is the set of nodes of T in depth k .

in this BSC channel we can see that for a connected $i \rightarrow j$ we have

$$P[X_i = X_j] = 1 - \delta \quad \& \quad e = (i, j) \rightarrow X_i \oplus X_j = Z_e \sim \text{Ber}(\delta)$$

$$\Rightarrow Y_e = Z_e \oplus Z_e = 0 \quad \forall e \in E$$

So in this case which we can see that Y_e is always zero

So it has no uncertainty & we can drop it and nothing will happen

from previous section $\rightarrow I(X_p; X_{S_k}, \underbrace{Y_e}_{\text{drop this}}) = I(X_p; X_S) \leq \underbrace{P_{(S, n)}[P \rightarrow S_k]}_1 \log 2 = \log 2$
 (not useful)

$$|S_k| = (d+1)^k \rightarrow d_{TV}(X_p, X_{S_k}) = \frac{1}{|S_k|} \sum_{i \in S_k} d_{TV}(X_p, X_i)$$

Let $V(p, i)$ be the set of all nodes connecting $p \rightarrow i$ in order. we have:

$$d_{TV}(X_p, X_i) \leq \sum_{j \in V(p, i)} d_{TV}(X_p, X_j)$$

ideas way to relate d_{TV} to I from both sides

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(I) let g to be a non-decreasing function, then

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$$\mathbb{P}\left[\frac{dV}{d\mu} > M\right] \leq \mathbb{P}\left[g\left(\frac{dV}{d\mu}\right) > g(M)\right]$$

$$\begin{aligned} \mathbb{P}\left(\frac{dV(x)}{d\mu(x)} > M\right) &= \mathbb{E}_\mu \left[\mathbb{1}\left[\frac{dV(x)}{d\mu(x)} > M\right] \right] \xrightarrow{\text{change of measure}} \mathbb{E}_\mu \left[\frac{dV}{d\mu} \cdot \mathbb{1}\left[\frac{dV}{d\mu} > M\right] \right] \\ &\xrightarrow{\text{let } t \rightarrow} = \int_0^\infty \mu \left[\frac{dV(x)}{d\mu(x)} > t \right] dt \end{aligned}$$

using the lemma given in the group!

$$= \int_M^\infty \mu \left[\frac{dV(x)}{d\mu(x)} > t \right] dt \leq \int_M^\infty \mu \left[f\left(\frac{dV(x)}{d\mu(x)}\right) > f(t) \right] dt$$

$$\Rightarrow \mathbb{P}\left[\frac{dV}{d\mu}(x) > M\right] \leq \int_M^\infty \mu \left[f\left(\frac{dV}{d\mu}\right) > f(t) \right] dt \xrightarrow{\text{change of variable}} \int_{f(M)}^\infty \mu \left[f\left(\frac{dV}{d\mu}\right) > u \right] \frac{du}{f'(u)}$$

since $f'(t) \geq f'(M)$

$$\mathbb{P}\left[\frac{dV}{d\mu}(x) > M\right] \leq \frac{1}{f'(M)} \int_{f(M)}^\infty \mu \left[f\left(\frac{dV}{d\mu}(x) > u \right] du = \frac{\mathbb{E}_\mu \left[f\left(\frac{dV}{d\mu}\right) \right]}{f'(M)}$$

(II)

$$\leq \log(1 + \delta^2 (\mathbb{P}_{Y|E} \parallel \mathbb{Q}_Y))$$

$$D_{KL}(\mathbb{P}_Y \parallel \mathbb{Q}_Y) = D_{KL}(\bar{\delta} \mathbb{P}_{Y|E} + \delta \mathbb{P}_{Y|E} \parallel \mathbb{Q}_Y) \leq \bar{\delta} D_{KL}(\mathbb{P}_{Y|E}) + \delta D_{KL}(\mathbb{P}_{Y|E} \parallel \mathbb{Q}_Y)$$

writing the KL divergence as a probability:

$$\mathbb{P}_{Y|X, E^c}(y) = \frac{\mathbb{P}_{Y, E^c}(x|y)}{\mathbb{P}(E^c)} = \frac{\mathbb{1}\{y \in E^c\} \mathbb{P}_{Y|X}(y)}{\delta}$$

$$\leq \mathbb{E}_Y [D_{KL}(\mathbb{P}_{Y|X, E^c} \parallel \mathbb{Q}_Y)]$$

$$D_{KL}(\mathbb{P}_{Y|X, E^c} \parallel \mathbb{Q}_Y) = \sum_{y \in Y} \frac{\mathbb{1}(y \in E^c)}{\delta} \mathbb{P}_{Y|X}(y) \log \left(\frac{\mathbb{P}_{Y|X}(y)}{\delta \mathbb{Q}(y)} \right) = \mathbb{E}_{\mathbb{P}_{Y|X}} \left[\frac{\mathbb{1}(y \in E^c)}{\delta} (-\log \delta) + \dots \right]$$

$$D_{KL}(\mathbb{P}_{Y|X, E^c} \parallel \mathbb{Q}_Y) \leq \bar{\delta} D_{KL}(\mathbb{P}_{Y|E} \parallel \mathbb{Q}_Y) + \log \delta \mathbb{E}_{X, Y} \left[\frac{\mathbb{1}(y \in E^c)}{\delta} \right] + \mathbb{E}_{X, Y} \left[\log \left(\frac{\mathbb{P}_{Y|X}}{\mathbb{Q}_Y} \right) \right]$$

Now we'll show that this second & third term below is $\sqrt{\delta \text{Var}(\dots)}$

$$\mathbb{E}_{X, Y} \left[\mathbb{1}(y \in E^c) \cdot \log \frac{\mathbb{P}_{Y|X}(y)}{\mathbb{Q}_Y(y)} \right] + \delta \mathbb{E}_X [D_{KL}(\mathbb{P}_{Y|X} \parallel \mathbb{Q}_Y)] - \mathbb{E} \left[\log \left(\frac{\mathbb{P}_{Y|X}}{\mathbb{Q}_Y} \right)^2 \right]$$

Now using the Cauchy-Schwarz inequality we can deduce: $\delta \leq$ the L.H.S. $\leq \sqrt{\mathbb{E} \left\{ \mathbb{I}(y \in \mathcal{C})^2 \right\}} \sqrt{\mathbb{E} \left[\log \left(\frac{P_{Y|X}}{Q_Y} \right) - \mathbb{E} \log \left(\frac{P_{Y|X}}{Q_Y} \right) \right]^2}$

now it implies that: $\delta = \sqrt{\delta} \cdot \sqrt{\text{Var} \left(\log \frac{P_{Y|X}}{Q_Y} \right)}$ (c.t.d.)

$$D_{KL}(P_Y \| Q_Y) \leq \log(1 + \delta^2 (P_{Y|X} \| Q_Y)) + \delta \log \delta + \delta \mathbb{E}_X \left[D_{KL}(P_{Y|X} \| Q_Y) \right] + \sqrt{\delta} \sqrt{\text{Var} \left(\log \left(\frac{P_{Y|X}}{Q_Y} \right) \right)}$$

* $X \rightarrow Y = \sqrt{\delta} X + Z$ (I) $z \sim N(0,1)$ let $\ell(X, \hat{x}) = (x - \hat{x})^2$ $\hat{x} = f(y)$ $\frac{4 \delta^2}{\text{Cov}(X, Y)}$ $\frac{Q.E.D.}{\text{Cov}(X, Y)}$

$$\ell(X, \hat{x}) = \mathbb{E}[(X - \hat{x})^2] = \mathbb{E}[(X - \mathbb{E}[X|Y] + \mathbb{E}[X|Y] - f(Y))^2] = \mathbb{E}_{X,Y}[(X - \mathbb{E}[X|Y])^2] + \mathbb{E}_{X,Y}[(\mathbb{E}[X|Y] - f(Y))^2] + 2\mathbb{E}_{X,Y}[(X - \mathbb{E}[X|Y])(\mathbb{E}[X|Y] - f(Y))]$$

$$\textcircled{C} \mathbb{E}_{X,Y}[(X - \mathbb{E}[X|Y])(\mathbb{E}[X|Y] - f(Y))] = \mathbb{E}_Y[(\mathbb{E}[X|Y] - f(Y)) \underbrace{\mathbb{E}_{X,Y}[X - \mathbb{E}[X|Y]]}_0]$$

$$\hookrightarrow \ell(X, \hat{x}) = \mathbb{E}[(X - \mathbb{E}[X|Y])^2] + \mathbb{E}[(\mathbb{E}[X|Y] - f(Y))^2] \quad \leftarrow \text{let to minimize}$$

the first term is not up to us how ever, the second term can be optimized to zero. thus $f(y) = \mathbb{E}[X|Y] = \arg \min_{f(y)} \ell(X, f(y))$.

* $\textcircled{II} Y = \sqrt{\delta} X + Z \rightarrow I(X; Y) = I(X; \sqrt{\delta} X + Z) = I(X; X + \sqrt{\delta} Z)$

$$I(X; Y) = \mathbb{E}_X [D_{KL}(P_{Y|X} \| P_Y)] - D_{KL}(P_Y \| P_Z)$$

$$= \mathbb{E}_X [D_{KL}(N(\sqrt{\delta} X, 1), N(0, 1))] - D_{KL}(P_Y \| P_Z)$$

$$= O\left(\frac{\delta^2}{2}\right)$$

$$= \mathbb{E}_X \left[\frac{\delta X^2}{2} \right] - D_{KL}(P_{\sqrt{\delta} X + Z} \| P_Z)$$

using Taylor series at $\delta=0$

$$D_{KL}(P_{\sqrt{\delta} X + Z} \| P_Z) = \delta \frac{d}{d\delta} D_{KL}(P_{\sqrt{\delta} X + Z} \| P_Z) \Big|_{\delta=0} + 0 + o(\delta^2)$$

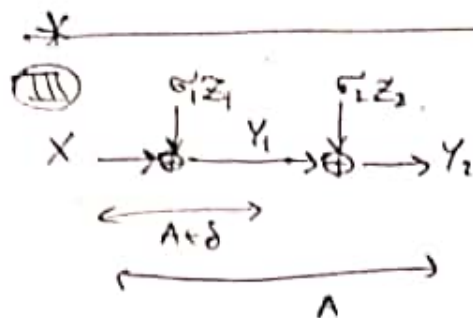
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Since both $Z \sim N(0,1)$ & $\sqrt{\delta}X + Z \sim N(\sqrt{\delta}X, 1)$ are normal distributions, we get

$$D_{KL}(P_{\sqrt{\delta}X+Z} \| P_Z) = \frac{1}{2} E[\delta X^2] + o(\delta) = \frac{1}{2} \delta (E[X^2]) + o(\delta)$$

$$\Rightarrow I(X; Y) = \frac{\delta}{2} [E[X^2] - E[X]^2] + o(\delta) \quad \text{Q.E.D.}$$

$\text{Var}(X) \rightarrow \text{equivalency to SNR}$



$$M_E(A) = E_{X,Y}[(X - E[X|Y])^2] = E_{X,Z}[(X - E[X|\sqrt{A}X + Z])^2]$$

Since the Markov kernel is of the form $X \rightarrow Y_1 \rightarrow Y_2$ & the fact that $X \perp\!\!\!\perp Y_2 | Y_1$ we can deduce that $I(X; Y_2 | Y_1) = 0$, thus:

$$I(Y_1, Y_2; X) = I(X; Y_1) + \underbrace{I(X; Y_2 | Y_1)}_{\text{conditioning on } Y_2} = I(X; Y_2) + I(X; Y_1 | Y_2) \quad (*)$$

$$\left. \begin{array}{l} I(X; Y_1) = I(A+\delta) \\ I(X; Y_2) = I(A) \end{array} \right\} \Rightarrow \frac{M_E(A)}{2} = \lim_{\delta \rightarrow 0} \frac{I(X; Y_1) - I(X; Y_2)}{\delta} = \lim_{\delta \rightarrow 0} \frac{I(A+\delta) - I(A)}{\delta}$$

so it suffices to prove that $\uparrow$ to assume $I(X; Y_1) - I(X; Y_2) = \delta \left(\frac{1}{2} M_E(A) + o(1) \right)$

$$IV) Y_1 = X + \sigma_1 Z_1 = X + \frac{Z_1}{\sqrt{A+\delta}}, \quad A = \frac{1}{\sigma_1^2 + \sigma_2^2} \quad \leftarrow \text{کار دو کدیر ایستادن منظم کردن}$$

$$Y_2 = X + \sigma_1 Z_1 + \sigma_2 Z_2 = X + \frac{Z_2}{\sqrt{A}}$$

$$(A+\delta)Y_1 = (A+\delta)X + \sqrt{A+\delta} Z_1 = \delta X + AX + \frac{Z_1}{\sigma_1} = \delta X + \frac{AY_2}{\sqrt{\sigma_1^2 + \sigma_2^2}} \cdot \left(\frac{\sigma_1^2 Z_1 + \sigma_2^2 Z_2}{\sqrt{\sigma_1^2 + \sigma_2^2}} \right)$$

$$\text{which results in } (A+\delta)Y_1 = AY_2 + \delta X + \frac{Z_1(\sigma_1^2 + \sigma_2^2) - \sigma_1^2 Z_1 - \sigma_1 \sigma_2 Z_2}{\sigma_1(\sigma_1^2 + \sigma_2^2)} = AY_2 + \delta X$$

$$\left. \begin{array}{l} \text{so by } A+\delta = \frac{1}{\sigma_1^2} \\ A = \frac{1}{\sigma_1^2 + \sigma_2^2} \end{array} \right\} \text{ it will be } \Rightarrow \delta = \frac{\sigma_2^2}{\sigma_1^2(\sigma_1^2 + \sigma_2^2)}$$

So we get $(A+\delta)Y_1 = AY_2 + \delta X + \sqrt{\delta}Z$, $Z \perp X$, $Z \sim N(0,1)$ $\Rightarrow \sigma_1^2 Z_1 - \sigma_2^2 Z_2$
 $\sqrt{\sigma_1^2 + \sigma_2^2}$

so directly $\begin{cases} (A+\delta)Y_1 = AY_2 + \delta X + \sqrt{\delta}Z \\ Z \perp X \sim N(0,1) \end{cases}$ ~~Q.E.D.~~ Q.E.D.

* Q.E.D.

Ⓐ Now directly, we will show that $I(X; Y_1 | Y_2) = \frac{\delta}{2} M_E(A) + o(1)$

to show: we have to bound $I(X; Y_1) - I(X; Y_2)$

$$I(X; Y_1) - I(X; Y_2) = I(X; Y_1 | Y_2) = \mathbb{E}_{Y_2, X} \left[D_{KL}(P_{Y_1 | Y_2, X} \| P_Z) \right]$$

$$\left. \begin{aligned} P_Z &\rightarrow \hat{Z} = \frac{\sqrt{\delta}Z + \delta \mathbb{E}X | Y_2 + AY_2}{A+\delta} \\ Y_1 &= Y_2 - \sigma_2 Z \\ \sigma_2^2 &= \frac{1}{A} - \frac{1}{A+\delta} \end{aligned} \right\} \text{so we will get:}$$

they differ in X and $\mathbb{E}X | Y_2$:

$$I(X; Y_1 | Y_2) = \mathbb{E}_{Y_2} [I(X; Y_1 | Y_2)] = \mathbb{E}_{Y_2, X} \left[D_{KL} \left(P_{\frac{AY_2 + \delta Z + \delta X}{A+\delta}} \| P_{\frac{AY_2 + \sqrt{\delta}Z + \delta \mathbb{E}X | Y_2}{A+\delta}} \right) \right]$$

$$= \mathbb{E}_{Y_2} \left[D_{KL} \left(P_{\frac{AY_2 + \sqrt{\delta}Z + \delta X}{A+\delta} | Y_2} \| P_{\frac{AY_2 + \sqrt{\delta}Z + \delta \mathbb{E}X | Y_2}{A+\delta} | Y_2} \right) \right]$$

So the first term will be:

$$\mathbb{E}_{Y_2, X} \left[D_{KL} \left(\mathcal{N} \left(\frac{AY_2 + \delta X}{A+\delta}, \frac{\delta}{(A+\delta)^2} \right) \| \mathcal{N} \left(\frac{AY_2 + \delta \mathbb{E}X | Y_2}{A+\delta}, \frac{\delta}{(A+\delta)^2} \right) \right) \right]$$

$$= \frac{(\lambda_2 - \lambda_1)^2}{2} = \frac{\delta}{2} \cdot \frac{\mathbb{E}[(X - \mathbb{E}X | Y_2)^2]}{A+\delta} = \frac{\delta}{2} M_E(A)$$

and the second term will be $o(1)$, so it will imply that:

$$I(X; Y_1 | Y_2) = \frac{\delta}{2} M_E(A) + o(1) \text{ which proves the whole theorem.}$$

Q.E.D.

① $P_\pi = \int P_{\theta\pi}(d\theta) \quad W(\theta, \hat{\theta}) = \mathbb{E}_\pi \left[\frac{P_\theta P_{\hat{\theta}}}{q^2} \right]$

from the definition of χ^2 we can say that $\chi^2(P_\pi \| Q) = \mathbb{E}_{\theta, \hat{\theta} \sim \pi} [W(\theta, \hat{\theta})] - 1$
which we'll prove!

$$\rightarrow \mathbb{E}_{\theta, \hat{\theta} \sim \pi} [W(\theta, \hat{\theta})] = \mathbb{E}_{\theta, \hat{\theta} \sim \pi} \left[\mathbb{E}_Q \left[\frac{P_\theta P_{\hat{\theta}}}{q^2} \right] \right] = \mathbb{E}_Q \left[\mathbb{E}_{\theta \sim \pi} \left[\mathbb{E}_{\hat{\theta} \sim \pi} \frac{P_\theta P_{\hat{\theta}}}{q^2} \right] \right]$$

change the order of expected $\mathbb{E}$

Since $\mathbb{E}_\theta [P_\theta] = P_\pi = \mathbb{E}_{\hat{\theta}} [P_{\hat{\theta}}]$ we can deduce that:

$$\mathbb{E}_{\theta, \hat{\theta} \sim \pi} [W(\theta, \hat{\theta})] = \mathbb{E}_Q \left[\frac{P_\pi^2}{q^2} \right] = -1 + 1 + \mathbb{E}_Q \left[\frac{P_\pi^2}{q^2} \right] = 1 + \mathbb{E}_Q \left[\frac{P_\pi^2 - q^2}{q^2} \right] \quad \chi^2(P_\pi, Q)$$

so therefore: $\chi^2(P_\pi \| Q) = \mathbb{E}_{\theta, \hat{\theta}} \left[\frac{P_\theta P_{\hat{\theta}}}{q^2} \right] - 1$

② $P = \{P_n\}_{n=1}^\infty$

$Q = \{Q_n\}_{n=1}^\infty \quad \chi^2(P_n \| Q_n) = \alpha_n$

$$\infty > \|L_n\|^2 = \mathbb{E}_{X \sim Q_n} \left[\frac{P_n(X)^2}{Q_n(X)^2} \right] = 1 + \chi^2(P_n \| Q_n) < \infty$$

so as we saw, we can deduce that if $\forall n: \chi^2(P_n \| Q_n) < \infty$ then $\chi^2(P_n \| Q_n) = \alpha_n$ which leads to $P \triangleleft Q$. Q.E.D.

① $\sigma \in \{1, -1\}^n$, $A_{ij} = \begin{cases} P & \sigma_i = \sigma_j \\ Q & \sigma_i \neq \sigma_j \end{cases} \rightarrow \text{the Adjacency Matrix}$

$A = (A_{ij})_{n \times n} \quad \begin{matrix} P \rightarrow \text{Ber}(P) \\ Q \rightarrow \text{Ber}(Q) \end{matrix} \leftrightarrow G = G(\sigma, P, Q)$

Hypothesis test $\rightarrow \begin{cases} H_0: G \stackrel{\text{i.i.d.}}{\sim} R_0 = G(n, \frac{P+Q}{2}) \leftrightarrow \text{all from } \frac{P+Q}{2} \\ H_1: G \stackrel{\text{i.i.d.}}{\sim} R_1 = G(P, Q) \end{cases}$

$$P(A) = \prod_{1 \leq i < j \leq n} P(A_{ij})^{\mathbb{1}(\sigma_i = \sigma_j)} \cdot \prod_{1 \leq i < j \leq n} Q(A_{ij})^{\mathbb{1}(\sigma_i \neq \sigma_j)}$$

the distribution of A matrix

(which is symmetric)

since $\sigma_i \in \{\pm 1\}$ we can deduce that for each Rademacher variable we have

$$\begin{cases} \mathbb{1}\{\sigma_i = \sigma_j\} = \frac{\sigma_i \sigma_j + 1}{2} \in \{0, 1\} \\ \mathbb{1}\{\sigma_i \neq \sigma_j\} = \frac{-\sigma_i \sigma_j + 1}{2} \in \{0, 1\} \end{cases}$$

$$P(A) = \prod_{1 \leq i < j \leq n} \sqrt{P(A_{ij})q(A_{ij}) \left(\frac{P(A_{ij})}{q(A_{ij})} \right)^{\sigma_i \sigma_j}}$$

now let $P(A)$ be the distribution for $R = \frac{P+Q}{2}$ distribution we have

$$R(A) = \prod_{1 \leq i < j \leq n} \frac{P(A_{ij}) + q(A_{ij})}{2}$$

Now with the idea of Problem 5 to generate $\frac{P_0 P_0^*}{q^2}$ we get:

$$\frac{P(A) P_0^*(A)}{R(A)^2} = \prod_{1 \leq i < j \leq n} \frac{4 P(A_{ij}) q(A_{ij})}{(P(A_{ij}) + q(A_{ij}))^2} \cdot \left(\frac{P(A_{ij})}{q(A_{ij})} \right)^{\frac{\sigma_i \sigma_j + \hat{\sigma}_i \hat{\sigma}_j}{2}}$$

$$\text{Now let } W(\underline{\sigma}, \underline{\hat{\sigma}}) = \mathbb{E}_{A_{ij} \sim R} \left[\frac{P_0(A) P_0^*(A)}{R(A)^2} \right] = \prod_{1 \leq i < j \leq n} \mathbb{E}_{A_{ij} \sim R} \left[\frac{P_0(A_{ij}) P_0^*(A_{ij})}{R(A_{ij})^2} \right]$$

finally $\rightarrow W(\underline{\sigma}, \underline{\hat{\sigma}}) = \prod_{1 \leq i < j \leq n} \int_{\mathcal{X}} \frac{2 P(x) q(x)}{[P(x) + q(x)]^2} \left(\frac{P(x)}{q(x)} \right)^{\frac{\sigma_i \sigma_j + \hat{\sigma}_i \hat{\sigma}_j}{2}} dx$

$$= \prod_{1 \leq i < j \leq n} \int_{\mathcal{X}} \frac{1}{(P(x) + q(x))} \times P(x)^{\mathbb{1}(\sigma_i = \sigma_j)} \cdot q(x)^{\mathbb{1}(\sigma_i \neq \sigma_j)} \cdot P(x)^{\mathbb{1}(\hat{\sigma}_i = \hat{\sigma}_j)} \cdot q(x)^{\mathbb{1}(\hat{\sigma}_i \neq \hat{\sigma}_j)} dx$$

Now that we got here we can rewrite $P \times q$ as the sum of the phrases

$$\begin{aligned} \Rightarrow P(x)^{\mathbb{1}(\sigma_i = \sigma_j)} \cdot q(x)^{\mathbb{1}(\sigma_i \neq \sigma_j)} &= \mathbb{1}(\sigma_i = \sigma_j) P(x) + \mathbb{1}(\sigma_i \neq \sigma_j) q(x) \\ &= \frac{P(x) + q(x)}{2} + \sigma_i \sigma_j \frac{P(x) - q(x)}{2} \end{aligned}$$

thus:

$$\Rightarrow W(\underline{\sigma}, \underline{\hat{\sigma}}) = \prod_{1 \leq i < j \leq n} \int_{\mathcal{X}} \frac{2}{P(x) + q(x)} \left[\frac{P(x) + q(x)}{2} + \frac{\sigma_i \sigma_j (P(x) - q(x))}{2} \right] \times \left[\frac{P(x) + q(x)}{2} + \frac{\hat{\sigma}_i \hat{\sigma}_j (P(x) - q(x))}{2} \right] dx$$

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$$\Rightarrow W(\underline{\sigma}, \underline{\hat{\sigma}}) = \prod_{1 \leq i < j \leq n} \int_{\mathcal{X}} \left(\frac{2}{P(x) + q(x)} \right) \left[(P(x) + q(x))^2 + \sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j (P(x) - q(x))^2 + (\sigma_i \sigma_j + \hat{\sigma}_i \hat{\sigma}_j) (P(x) + q(x))^2 \right] dx$$

& after simplification we will get

$$W(\underline{\sigma}, \underline{\hat{\sigma}}) = \prod_{1 \leq i < j \leq n} \frac{1}{2} \int \left[p(x_1) + q(x_1) + (\sigma_i \sigma_j + \hat{\sigma}_i \hat{\sigma}_j) (p(x_1) - q(x_1)) + \sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j \frac{(p(x_1) - q(x_1))^2}{p(x_1) + q(x_1)} \right] dx$$

now since we are asked/given $\rho = \int \frac{(p(x) - q(x))^2}{2(p(x) + q(x))} dx$ we get:

$$W(\underline{\sigma}, \underline{\hat{\sigma}}) = \prod_{1 \leq i < j \leq n} \left[\frac{1+1}{2} + \underbrace{\left(\frac{\sigma_i \sigma_j + \hat{\sigma}_i \hat{\sigma}_j}{2} \right)}_0 (1-1) + \sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j \rho \right]$$

$$= \prod_{1 \leq i < j \leq n} \left[1 + \sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j \rho \right] \leq \prod_{i < j} e^{\sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j \rho}$$

$$\Rightarrow W(\underline{\sigma}, \underline{\hat{\sigma}}) = \exp \left(\sum_{1 \leq i < j \leq n} \hat{\sigma}_i \hat{\sigma}_j \sigma_i \sigma_j \rho \right) = \exp \left(\frac{\rho}{2} \sum_{i,j=1}^n (\sigma_i \sigma_j \hat{\sigma}_i \hat{\sigma}_j) \right)$$

$$= \exp \left(\frac{\rho}{2} \sum_{i=1}^n \sigma_i \hat{\sigma}_i \left(\sum_{j=1}^n \sigma_j \hat{\sigma}_j \right) \right) = \exp \left(\frac{\rho}{2} \langle \underline{\sigma}, \underline{\hat{\sigma}} \rangle^2 \right)$$

Q.E.D. ■

Ⓐ Now in this part we'll investigate if $p = \frac{a}{n}, q = \frac{b}{n}, \tau = \frac{(a-b)^2}{2(a+b)}$ what will be $\rho = \frac{\tau + o(1)}{n}$. So we compute ρ :

$$\rho = \frac{1}{2} \int \frac{(p(x_1) - q(x_1))^2}{p(x_1) + q(x_1)} dx = \frac{(p-q)^2}{2(p+q)} + \frac{(\bar{p}-\bar{q})^2}{2(\bar{p}+\bar{q})} = \frac{(p-q)^2}{(p+q)(\bar{p}+\bar{q})}$$

$\begin{cases} p(x_1) \sim \text{Ber}(p) \\ q(x_1) \sim \text{Ber}(q) \end{cases}$

which finally implies that: $\rho = \frac{(p-q)^2}{(p+q)(2-p-q)}$ $\xrightarrow[p=\frac{b}{n}]{p=\frac{a}{n}}$

$$\Rightarrow \rho = \frac{\left(\frac{a-b}{n}\right)^2}{2\left(\frac{a+b}{n}\right) - \left(\frac{a+b}{n}\right)^2} = \frac{1}{n} \cdot \frac{(a-b)^2}{2(a+b)} \times \underbrace{\left(\frac{1}{1 - \frac{a+b}{2n}} \right)}_{\sim 1 + \frac{a+b}{2n} + \dots}$$

$$= \frac{\tau}{n} \times [o(1) + 1] = \rho \quad \text{Q.E.D.} \quad \blacksquare$$

III Since $\sigma_i, \hat{\sigma}_i \in \text{Radmacher} \rightarrow \sigma_i \hat{\sigma}_i \in \text{Radmacher}$ so by symmetry we expect their inner product to be zero $\rightarrow \mathbb{E}[\langle \sigma_i, \hat{\sigma}_i \rangle] = 0$

so we want to bound $\chi^2(R_0 \| R_1)$ to see when we can't distinguish the parameters;

$$\chi^2(R_0 \| R_1) \leq \mathbb{E}_{\sigma_i \hat{\sigma}_i \sim \text{Radmacher}} \left(\lim_{n \rightarrow \infty} \exp \left(\frac{\langle \sigma_i, \hat{\sigma}_i \rangle^2}{n} \cdot \frac{\tau + o(1)}{2} \right) - 1 \right)$$

let $H = \langle \sigma_i, \hat{\sigma}_i \rangle$ since $\mathbb{E}[H] = 0 \rightarrow \mathbb{E}[H^2] = \text{Var}(H) = S^2 = \overline{H^2}$

so by CLT we get that

$$\chi^2(R_0 \| R_1) \leq \mathbb{E}_{S \sim N(0,1)} \left[\exp \left(\frac{\tau}{2} S^2 \right) \right] - 1 = -1 + \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{\frac{\tau}{2} s^2 - \frac{s^2}{2}} ds$$

$\downarrow$
 $\mathbb{E}[H^2]$

$$\chi^2(R_0 \| R_1) \leq -1 + \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\frac{s^2}{2}(1-\tau)} ds \rightarrow \text{now it can easily be seen}$$

that if $1-\tau \leq 0$ then the coeff of $-\frac{s^2}{2}$ in the exponent will be greater equal to zero & that makes the integral to diverge.

so we'll get:

$$\chi^2(R_0 \| R_1) \leq \begin{cases} 1 + \frac{1}{\sqrt{1-\tau}} & \tau < 1 \\ 1 + \infty & \tau \geq 1 \end{cases}$$

so we now assume that $\tau < 1$

so $\chi^2(R_0 \| R_1) = \frac{1}{\sqrt{1-\tau}} - 1 = o(1)$ and we can distinguish

$\chi^2(R_0 \| R_1)$ if $\chi^2 < \infty$.

However if $\tau \geq 1$ then $\chi^2(R_0 \| R_1)$ can be anything since $\chi^2 < \infty$ and we can't do anything in this case

$$\tau = \frac{(a-b)^2}{2(a+b)} < 1$$

Q.E.D.

① As it was discussed in the class, all $\mathcal{P}$ -divs are convex. let $\mathcal{R} = \mathcal{P}(\mathcal{D})$

P_1, Q_1, P_2, Q_2 be some distributions when $(D_f(P_1, Q_1), D_g(P_1, Q_1)) \in \mathcal{R}$
 $(D_f(P_2, Q_2), D_g(P_2, Q_2)) \in \mathcal{R}$: then $\forall \lambda \in [0, 1]$ any convex hull of

$(P_1, P_2, Q_1, Q_2) : (\lambda P_1 + \bar{\lambda} P_2)$ and $(\lambda Q_1 + \bar{\lambda} Q_2)$ are distributions.

thus $\Rightarrow (D_f(\lambda P_1 + \bar{\lambda} P_2, \lambda Q_1 + \bar{\lambda} Q_2), D_g(\lambda P_1 + \bar{\lambda} P_2, \lambda Q_1 + \bar{\lambda} Q_2)) \in \mathcal{R}$

we also have

$$D_f(\lambda P_1 + \bar{\lambda} P_2, \lambda Q_1 + \bar{\lambda} Q_2) \leq \lambda D_f(P_1, Q_1) + \bar{\lambda} D_f(P_2, Q_2)$$

$$D_g(\lambda P_1 + \bar{\lambda} P_2, \lambda Q_1 + \bar{\lambda} Q_2) \leq \lambda D_g(P_1, Q_1) + \bar{\lambda} D_g(P_2, Q_2)$$

so since between to points of distributions their convex hull will also be in the so called $\mathcal{R}$, we can prove that it's convex



② $D_f(P||Q) = \mathbb{E}_{x \sim Q} \left[f\left(\frac{dP}{dQ}\right) \right] = \mathbb{E}_{x \sim P} \left[\frac{dQ}{dP} f\left(\frac{dP}{dQ}\right) \right] = \mathbb{E}_{x \sim P} \left[\tilde{f}\left(\frac{dQ}{dP}\right) \right] = D_{\tilde{f}}(\tilde{Q}||\tilde{P})$
 $\tilde{f}(1/t) = t f(t)$

now let $Z \sim Q, H \sim R, H = \frac{P(Z)}{Q(Z)}$

$$D_f(P||Q) = \sum_{x \in X} q(x) f\left(\frac{p(x)}{q(x)}\right) = \sum_{x: q(x) > 0} q(x) f\left(\frac{dP}{dQ}\right) + \sum_{x: q(x) = 0} p(x) f\left(\frac{dP}{dQ}\right)$$

$$= \sum_{x: q(x) > 0} q(x) f\left(\frac{dP}{dQ}\right) + \sum_{x: q(x) = 0} p(x) f\left(\frac{dP}{dQ}\right)$$

by change of measure $f \rightarrow \tilde{f}$
 $\tilde{f}(1) = P(Q=1) = 1$
 $= \tilde{f}(1) (1 - P(Q \neq 1)) = \tilde{f}(1) (1 - \sum_{x \neq 1} q(x) \frac{p(x)}{q(x)}) = 1 - \mathbb{E}_{x \sim Q} \left[\frac{p(x)}{q(x)} \right] = 1 - \mathbb{E}_R[H]$

$\Rightarrow$ As a result: $\sum_{x \in X} q(x) f\left(\frac{p(x)}{q(x)}\right) = \mathbb{E}_{x \sim R} [f(x)]$

III) firstly, we can easily see that $\forall k \in \mathbb{N}$ $R_k \leq R_{k+1} \leq \mathbb{R}$: بالسهولة

So we can see that $P_k(P_{k+1}) = \mathbb{E}_{x \in R_k} [f(x)] + f(1) [1 - \mathbb{E}_{x \in R_k} [x]]$
 holds for both R and R_k since we didn't assume any
 binding assumptions.

→ Q.E.D. ■

III) Lenchel - Eggleston - Carathéodory

$S \subseteq \mathbb{R}^d$, $x \in \text{Co}(S)$, $\exists S' \subseteq \mathbb{R}^{d+1}$, $S' = \{x_1, \dots, x_{d+1}\}$, $S \rightarrow \text{info}$
 $x \in \text{Co}(S')$

→ $\exists \lambda, \alpha$ such that $\begin{pmatrix} \mathbb{E} x \\ \mathbb{E} f(x) \\ \mathbb{E} g(x) \end{pmatrix} = \alpha^T (\lambda, f(x), g(x))$

$\alpha = (\alpha_1, \alpha_2, \alpha_3)^T$
 $x = (x_1, x_2, x_3)^T$ since $\mathbb{E} x = \mathbb{E} x'$

→ $\begin{cases} \mathbb{E} f(x) + (1 - \mathbb{E} x) \tilde{f}(1) = \mathbb{E} f(x') + (1 - \mathbb{E} x') \tilde{f}(1) \\ \mathbb{E} g(x) + (1 - \mathbb{E} x) \tilde{g}(1) = \mathbb{E} g(x') + (1 - \mathbb{E} x') \tilde{g}(1) \end{cases} \rightarrow \in \mathbb{R}_1$

So

→ $\forall x \in R, \exists x' \in R_k \rightarrow x = x' \rightarrow R = R_k \rightarrow \text{Corollary } R_k \subset R_{k+1}$

IV) $E f(x) = \sum_{x \in X_k} P(x) f(x) = \sum_{i=1}^k P(x_i) f(x_i)$:70 b-201

Can be written as: $\sum_{i=1}^{k-1} P(x_i) f(x_i) + f(x_k) P(x_k)$

Let $P(x_i) = \frac{P(x_i)}{1 - P(x_k)}$ for normalization, then:

$$\sum_i P(x_i) = \frac{\sum P(x_i)}{1 - P(x_k)} = \frac{1 - P(x_k)}{1 - P(x_k)} = 1$$

For each random variable X with its support vector with k elements:
 $\exists X'$ with $(k-1)$ be its size of support vector & --
 random variable

-- (Let) $E f(x) = \alpha f(x_k) + (1-\alpha) E f(x')$

in this analogy x, x'' also take $(k-1)$ values: by connecting

$$\begin{pmatrix} E f(x) + (1-E f(x)) \tilde{f}(1) \\ E g(x) + (1-E f(x)) \tilde{g}(1) \end{pmatrix} = \alpha \begin{pmatrix} E f(x') + (1-E f(x')) \tilde{f}(1) \\ E g(x') + (1-E f(x')) \tilde{g}(1) \end{pmatrix} + \alpha \begin{pmatrix} E f(x'') + (1-E f(x'')) \tilde{f}(1) \\ E g(x'') + (1-E f(x'')) \tilde{g}(1) \end{pmatrix}$$

therefore each point in R_k was written by convex hull of points in R_{k-1} $\rightarrow$ therefore $\forall k: R_k \subseteq R_{k-1}$

$R_k \subseteq \text{Convex hull}(R_{k-1})$

V) so it was proven in the last part since $k=3$ will

result in $R = R_1 = CH(R_3) = CH(CH(R_2)) = CH(R_1) =$

Q.E.D ■